Structural and Multidisciplinary Optimization (2024) 67:44
https://doi.org/10.1007/s00158-024-03751-9

RESEARCH PAPER

A hybrid criterion-based sample infilling strategy for surrogate-assisted multi-objective optimization

Puyi Wang[1] · Yingchun Bai[1] · Cheng Lin[1] · Xu Han[2]

Received: 25 May 2023 / Revised: 4 October 2023 / Accepted: 21 January 2024 / Published online: 6 March 2024
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024

Abstract
High-quality Pareto front is always pursued when solving multi-objective optimization problems with surrogate-assisted multi-objective optimization (SMOO). For this purpose, a hybrid criterion-based sample infilling strategy is proposed to improve the predicted performance of surrogate model. Two infill criteria are integrated in proposed strategy, one is the expected improvement matrix-based infill criterion, by which a new sample point is generated through maximizing the expected improvement function of the non-dominated solutions in objective space. The other one is the Euclidean distance-based infill criterion, by which a newly different sample point is generated through perturbing the non-dominated solutions in design space. At each iteration in the SMOO process, two promising infilling sample points are obtained by the proposed strategy with the filter strategy, and the corresponding responses are further evaluated with the real simulation models, then the sample set and surrogate models will be updated sequentially. According to the results of the numerical validations, the SMOO algorithm with the proposed strategy has the competitive capabilities in terms of high-quality Pareto front and computational stability. Finally, the hybrid criterion-based sample infilling strategy for SMOO is applied to solve a high-dimension multi-objective optimization problem related to lightweight design of an electric bus body structure.

Keywords Multi-objective optimization · Surrogate model · Hybrid sample infilling · Expected improvement · Pareto front

1 Introduction

Surrogate models have been widely utilized to approximate multi-objective optimization problems in practical engineering for alleviating the expensive cost (Xu et al. 2022; Hou et al. 2014; Chugh et al. 2019). Since the approximation accuracy of surrogate model plays an important role in the optimum quality, how to improve the predicted accuracy and robustness has been a hot research topic in surrogate-assisted multi-objective optimization (SMOO) (Wang and Shan 2007). Typically, surrogate models are updated through sample infilling strategy during the optimization to improve the approximation and whereby optimization accuracy (Jin 2011).

Presently, the main infill criteria based on the expected improvement (EI) (Wagner et al. 2010), the low confidence bound (Liu et al. 2014), and the probability of improvement are widely used to generate infill samples to reconstruct surrogate models (Couckuyt et al. 2014; Horn et al. 2015). Generally, the low confidence bound-based criterion locates new sample point (sample point) based on a linear combination of the predicted value and the standard error. The probability of improvement-based criterion determines a new sample point in expected areas at which the objective function value has the maximum probability of improvement, while the EI-based criterion selects an additional sample point based on the maximum value of the expected improvement function. In practice, the low confidence bound-based criterion behaves poor robustness (Cheng et al. 2019), on the contrary, the EI-based criterion can provide the specific magnitude of improvement (Parr et al. 2012), thus it can be employed to obtain a more reliable infill sample for surrogate models reconstructing.

Responsible Editor: Axel Schumacher

✉ Yingchun Bai
baiyc@bit.edu.cn

1 National Engineeringngineering Research Center of Electriclectric Vehiclesehicles, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China

2 State Key Laboratory of Reliability and Intelligence of Electrical Equipment, Hebei University of Technology, Tianjin 300401, China

EI-based approach is originally developed for surrogate-assisted single-objective optimization, and then extended for the SMOO because of the efficient global optimization. For the single-objective optimization problem, Jones et al. (1998) constructed and maximized the EI function of the optimization objective to generate new infill sample points. Zhang et al. (2019) proposed a variable-fidelity EI maximization method to update multi-fidelity models adaptively. To further improve the distribution of extra samples, Zhang et al. (2021) developed an EI-based multi-peak parallel strategy to generate sample points within expected domains. By concentrating on the improvement of single optimization objective, Jeong et al. (2006) employed the EI-based infill criterion for newly additional sample points generating in SMOO. Unfortunately, it can generate extreme infill samples which leads to uneven Pareto front ultimately (Couckuyt et al. 2014). Thus, researchers have to develop alternative indicator-based sample infilling criteria for SMOO. Theoretically, the hypervolume indicator denotes the volume of the dominated region and estimates the convergence and uniformity of the non-dominated solutions. The larger the hypervolume value is, and the better performance of Pareto front is Zitzler and Thiele (1999). Due to this advantage, corresponding criteria with expected hypervolume improvement (EHVI) is developed for SMOO by many researchers.

In principle, the maximum value of EHVI function is at the newly additional sample point (Couckuyt et al. 2014). By calculating hypervolume value with Monte Carlo method (Bader and Zitzler 2011), Shimoyama et al. (2013) proposed an EHVI-based infill criterion to obtain additional samples for SMOO. To improve the computational efficiency, Li et al. (2018) introduced the conception of the local upper bounds to the evaluation of hypervolume for reducing the integral region (Klamroth et al. 2015), then proposed a modified EHVI sample infilling criterion to reduce the computational burden. Zhan et al. (2017) simplified the EHVI-based infill criterion to the cheap-to-evaluate expected improvement matrix (EIM), where each element of EIM represents an EI value of the non-dominated solution along the specified direction in objective space. Due to the effectiveness and high efficiency, the EIM-based criterion was used to solve the expensive simulation-based problem and the SMOO problem of engine calibration, respectively (He et al. 2021; Pal et al. 2021).

To improve the EHVI-based infilling efficiencies, multi-point infill criteria have been developed by improving the utilization potentiality of the historical data at a single optimization iteration. Li et al. (2021a, b) employed the EIM-based infill criterion and the reference vectors of the optimization objective to generate multiple sample points in one iteration. Yao and Yang (2021) presented an ensemble of four sub-infill criteria to obtain multiple sample points which are further screened by the filter strategy. While some researchers defined infill criteria from the other perspective. For example, Yi et al. (2014) developed a multi-point and multi-objective infill sampling criteria to get new sample points selected by picking some populations among non-dominated solutions. Namura et al. (2017) introduced the penalty-based boundary intersection to the EI-based criterion, then the sample points were uniformly obtained in the design space. Dong et al. (2019) presented a score-based multi-point infill criterion to determine additional sample points which were evaluated by the predicted performance-based scoring mechanism. Guo et al. (2021) proposed a filter-generalized EI infill criterion with considering the correlation of multi-fidelity models. Li et al. (2021a, b) proposed an efficient global optimization (EGO) method with multi-point infill sampling based on kriging through which multiple sample points can be obtained in each iteration by solving a multi-objective problem.

With the above infill criteria, the new infill samples are commonly obtained by evaluating the expensively real models. Nevertheless, in some cases, using virtual samples as the infill samples may be a cheap and feasible option to mitigate the expensive costs, because the virtual samples usually contain a certain potentially advantageous information despite of its low predicted accuracy (Wedyan et al. 2019). On this basis, some researchers have employed the virtual sample generation method to produce cheaper additional samples for the surrogate models rebuilding. Tang et al. (2020) employed the radial basis function-based surrogate model to generate abundant virtual samples, then further used the least squares support vector machine classifier to screen them. Li et al. (2013) employed the mega-trend-diffusion approach to generate high-quality virtual samples for updating the support vector regression-based surrogate model. Despite of the above successful appliances, to address practical engineering problems, we still expect newly high-quality samples to be added into sample set in SMOO. Except the predicted accuracy, the distribution and uniformity of the virtual samples should match to that of the true samples, but it is almost impossible for virtual samples based on the sparsely real data (Chen et al. 2021). The conditions of generating high-quality virtual samples are too stringent to be feasible for the expensive multi-objective optimization problems with sparse samples, therefore, the additional experiment and simulation data are necessary for reducing the uncertainty of the surrogate model (Jiang et al. 2016).

The above methods of infill samples can promote the qualities of the non-dominated solutions through exploring the objective space, but there is still systematic randomness which may lead to miss the suitable infill samples in SMOO (Gonzalez et al. 2020; Wu et al. 2021). To further improve the effectiveness of the additional samples, a hybrid criterion-based sample infilling strategy for surrogate-assisted multi-objective optimization is proposed, by which two infill

samples are obtained for reconstructing the surrogate models at each optimization iteration. As for the two infill samples, one infill sample obtained by the EIM-based infill criterion is for exploring the objective space, the other one obtained by the Euclidean distance-based infill criterion is for exploiting the design space.

The remainders are organized as follows. Theoretical fundamentals are briefly reviewed in Sect. 2. Then, the hybrid criterion-based sample infilling strategy and numerical validations are presented in Sects. 3 and 4, respectively. Furthermore, the proposed method is applied to the lightweight design of an electric bus body structure in Sect. 5, and some conclusions are drawn in Sect. 6.

2 Theoretical fundamentals

2.1 Multi-objective optimization problem

A multi-objective optimization problem can be generally modeled as

$$\begin{cases} Find\ \mathbf{x} = \left[x_1, x_2, \ldots, x_M\right] \\ \min\ \ f(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_N(\mathbf{x})] \\ s.t. \begin{cases} g_j(\mathbf{x}) \le 0, j = 1, 2, \ldots, J \\ h_k(\mathbf{x}) =, k = 1, 2, \ldots, K \\ x_i^L \le x_i \le x_i^U, i = 1, 2, \ldots, M \end{cases} \end{cases} \tag{1}$$

where x denotes an M-dimension vector of the design variable, M is the number of the design variables, $f(\mathbf{x})$ is the multi-objective vector which contains N objectives, $g_j(\mathbf{x})$ is the jth inequality constraint, and $h_k(\mathbf{x})$ is the kth equality constraint which can be transformed to the inequality forms by using a specific positive value. For the ith design variable, x_i^L is the lower bound, and x_i^U is the upper bound.

2.2 Kriging model

Based on the Gaussian process, the Kriging model applies the spatial interpolation technology to approximate optimization problems (Jones et al. 1998). The Kriging model can be expressed as

$$y(\mathbf{x}) = u(\mathbf{x}) + \varepsilon(\mathbf{x}) \tag{2}$$

where $y(\mathbf{x})$ is the predicted function value, and $u(\mathbf{x})$ is a regression function representing the mean value. $\varepsilon(\mathbf{x})$ is the error term and is defined as a Gaussian process following normal distribution $\mathrm{N}(0, \sigma^2)$.

Normally, a spatial correlation between the error terms of two points x_i and x_j is assumed, which is strongly related to their distance. By using the Gaussian function with a weighted distance, the correlation between x_i and x_j can be defined as

$$Corr(\varepsilon(x_i), \varepsilon(x_j)) = \exp\left(-\sum_{k=1}^{M} \theta_k (x_k^i - x_k^j)^{P_k}\right), \tag{3}$$

where $\theta_k\left(\theta_k \in (0, \infty)\right)$ is the kth element of M-dimension weight coefficient vector θ, which guarantees the anisotropy and accuracy of the constructed Kriging model. P_k is the exponential parameter ranging from 1 to 2, which is set to be 2 in this paper.

As a result, the Kriging model is determined by $2M+2$ super parameters, including $u, \sigma, \theta_1, \ldots, \theta_M, P_1, , \ldots, P_M$. These parameters are tuned by maximizing the likelihood of observed points. The Kriging model provides the predicted value of unobserved sample point x by

$$\hat{y}(\mathbf{x}) = \hat{u} + E^T C^{-1}(y - E\hat{u}), \tag{4}$$

and its associated mean squared error $S^2(\mathbf{x})$ is

$$S^2(\mathbf{x}) = \hat{\sigma}^2\left[1 - c^T C^{-1} c + \frac{1 - E^T C^{-1} c}{E^T C^{-1} E}\right], \tag{5}$$

where $\hat{y}(\mathbf{x})$ is the predicted value, $\mathbf{y} = (y_1, y_2, \ldots, y_n)$, is a vector containing n observed function values, $\boldsymbol{C}$ is the matrix with entry $C_{ij} = \mathrm{Corr}\left(\varepsilon\left(\mathbf{x}_i\right), \varepsilon\left(\mathbf{x}_j\right)\right)$, $\boldsymbol{c}$ is the n-dimension vector with the entry $c_i = \mathrm{Corr}\left(\varepsilon(\mathbf{x}), \varepsilon\left(\mathbf{x}_i\right)\right)$, and E is the n-dimension vector of ones.

2.3 Expected improvement-based infill criterion

To realize the global optimization, an EI-based infill criterion was proposed for updating the surrogate model by Jones et al. (1998). With this criterion, an infill sample point through maximizing the expected improvement is obtained at the unobserved sample points.

In Jones's approach, $Y(\mathbf{x})$ is the uncertainty response at unobserved sample point $\boldsymbol{x}$ following normal distribution

$$Y(\mathbf{x}) \sim \mathrm{N}(\hat{y}(\mathbf{x}), s^2(\mathbf{x})), \tag{6}$$

where $s(x)$ is the standard deviation. Then the improvement of $Y(x)$ is defined as

$$I(\mathbf{x}) = \max(y_{\min} - Y(\mathbf{x}), 0), \tag{7}$$

where $\mathrm{y}_{\min} = \min(y_1, y_2, \ldots, y_n)$,. Thus, the EI function is defined as

$$\mathrm{EI}(\mathbf{x}) = (y_{\min} - \hat{y}(\mathbf{x}))\Phi\left(\frac{y_{\min} - \hat{y}(\mathbf{x})}{s(\mathbf{x})}\right) + s(\mathbf{x})\phi\left(\frac{y_{\min} - \hat{y}(\mathbf{x})}{s(\mathbf{x})}\right), \tag{8}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the Gaussian cumulative distribution function and probability density function, respectively.

2.4 Expected improvement matrix-based infill criterion

To reduce the computational complexity of EHVI, Zhan et al. (2017) expanded the EI-based infill criterion into SMOO, and further proposed a 2-dimension EIM-based infill criterion to generate new infill samples for SMOO.

The EIM is expressed as

$$\begin{bmatrix} \mathrm{EI}_1^1(\mathbf{x}) & \cdots & \mathrm{EI}_N^1(\mathbf{x}) \\ \mathrm{EI}_1^2(\mathbf{x}) & \cdots & \mathrm{EI}_N^2(x) \\ \cdots & \ddots & \cdots \\ \mathrm{EI}_1^{N_{nds}}(\mathbf{x}) & \cdots & \mathrm{EI}_N^{N_{nds}}(\mathbf{x}) \end{bmatrix}, \tag{9}$$

where N_{nds} is the number of the non-dominated solutions. Relative to the jth non-dominated solution, $\mathrm{EI}_i^j = (\mathbf{x})$, is the EI value of the ith objective value at the unobserved sample point $\boldsymbol{x}$.

Since the modified hypervolume improvement can be expressed as

$$I_{hm}(\mathbf{x}) = \max_{j=1}^{N_{nds}} \left[\prod_{i=1}^{N} (r_i - y_i^j) - \prod_{i=1}^{N} \left(r_i - y_i(\mathbf{x}) \right) \right], \tag{10}$$

the EHVI-based infill criterion can be simplified as

$$\mathrm{EIM}_h(\mathbf{x}) = \max_{j=1}^{N_{nds}} \left[\prod_{i=1}^{N} (r_i - y_i^j) - \prod_{i=1}^{N} (r_i + \mathrm{EI}_i^j(\mathbf{x}) - y_i^j) \right], \tag{11}$$

where $y_i(\mathbf{x})$ is the ith objective response at the unobserved sample point $\boldsymbol{x}$. y_i^j is the ith objective of the jth non-dominated solution. r_i is the reference response of ith objective, and it is usually set to 1.1–1.2 times of maximum y_i^j. According to the Eq. (11), the $\mathrm{EIM}_h(\mathbf{x})$ is an infill criterion based on the hypervolume indicator.

3 Hybrid criterion-based sample infilling strategy for SMOO

3.1 Framework of the regular sample infilling strategy for SMOO

Generally, the framework of the regular sample infilling driven SMOO is provided in Fig. 1, and the computational details are as follows:

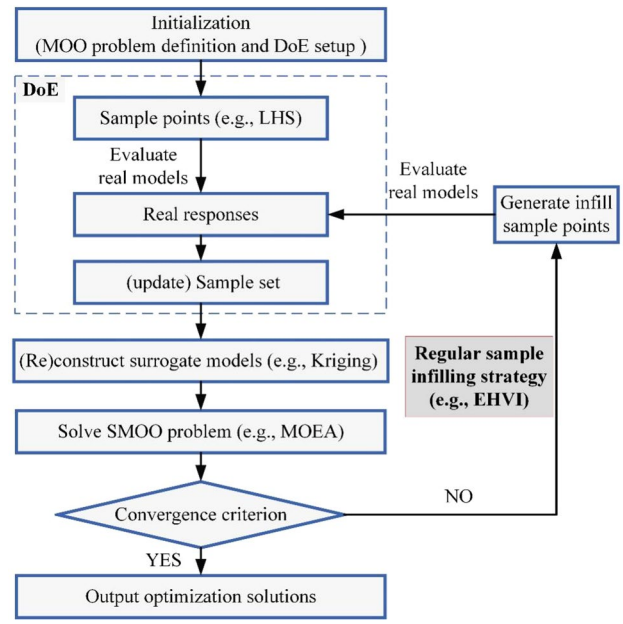


Fig. 1 Framework of the regular sample infilling for SMOO

Step 1 Initialization. The optimization problem and the design of experiment (DoE) setup are defined.

Step 2 Design of experiment. An experimental sample set is obtained by Latin hyper-cube sample (LHS) method generally, and the real responses are evaluated with the real simulation models at the sample points.

Step 3 Construction of the sample points. The sample points approximating the optimization problems are constructed with the real sample set.

Step 4 Solving the multi-objective optimization. The standard optimizer (e.g., multi-objective optimization (MOO) with evolutionary algorithm (MOEA)) is employed to solve the multi-objective optimization problem, and the non-dominated solutions are further obtained.

Step 5 Estimation of the convergence. If the convergence requirement is satisfied, go to the Step 7, otherwise, go to the Step 6 for generating new samples. The convergence criterion in SMOO is usually defined as the number of the total optimization iterations or the number of the real evaluation [6], and the latter will be selected in this paper.

Step 6 Generation of the infill samples. The new sample points are obtained by the sample infilling strategy (e.g., EHVI) and its responses are further gained by real evaluations.

Step 7 Output the final solutions.

3.2 Euclidean distance-based infill criterion

Essentially, the regular EHVI- and EIM_h-based infill criterion are developed for improving the quality of the Pareto front through exploring the objective space. Nevertheless, it is not always efficient to meet the requirement of the sample infilling. To overcome this disadvantage, the Euclidean distance (ED)-based infill criterion is proposed, by which the interesting infilling sample points will be further found out in promising deign space where the non-dominated solutions are contained within. In details, the EDs between each pair of the non-dominated solutions in design space are firstly calculated, then a penalized ED is defined, and the coordinates of the two non-dominated solutions with the maximum ED are perturbed for generating new sample points. Subsequently, the minimum EDs between the new sample points and the non-dominated solutions are calculated. On the above basis, a new infill candidate is defined through comparing the penalized ED with the minimum EDs, and the final infill sample point is generated by the filter strategy. Herein, the responses of the infill sample points are the real evaluations. For the clear description, the proposed infill criterion is illustrated through solving an unconstrained 2-dimension SMOO problem defined by the Eq. (1).

As Fig. 2 shows, the diagram of generating infill samples by the Euclidean distance-based infill criterion is illustrated, and an infilling sample generated by the EIM_h-based infill criterion is also shown. For clear illustration, $(X., F.)$ is the coordinates of the real samples, where 'X.' denotes the sample points in the design space, 'F.' denotes the responses at sample points in the objective space.

The Euclidean distance $d_{i,j}$ between a pair of sample points in design space is

$$d_{i,j} = \sqrt{\sum_{k=1}^{M} (x_{i,k} - x_{j,k})^2}, (i, j = 1, 2, \ldots, N_s. i \neq j), \tag{12}$$

where N_s is the number of the real samples. $x_{i,k}$ is the kth element of X_i, and $x_{j,k}$ is the kth element of X_j.

With the mutation factor δ, the penalized maximum difference of Euclidean distance d_p is defined as

$$d_p = \delta\left[\max(d_{i,j}) - \min(d_{i,j})\right], \tag{13}$$

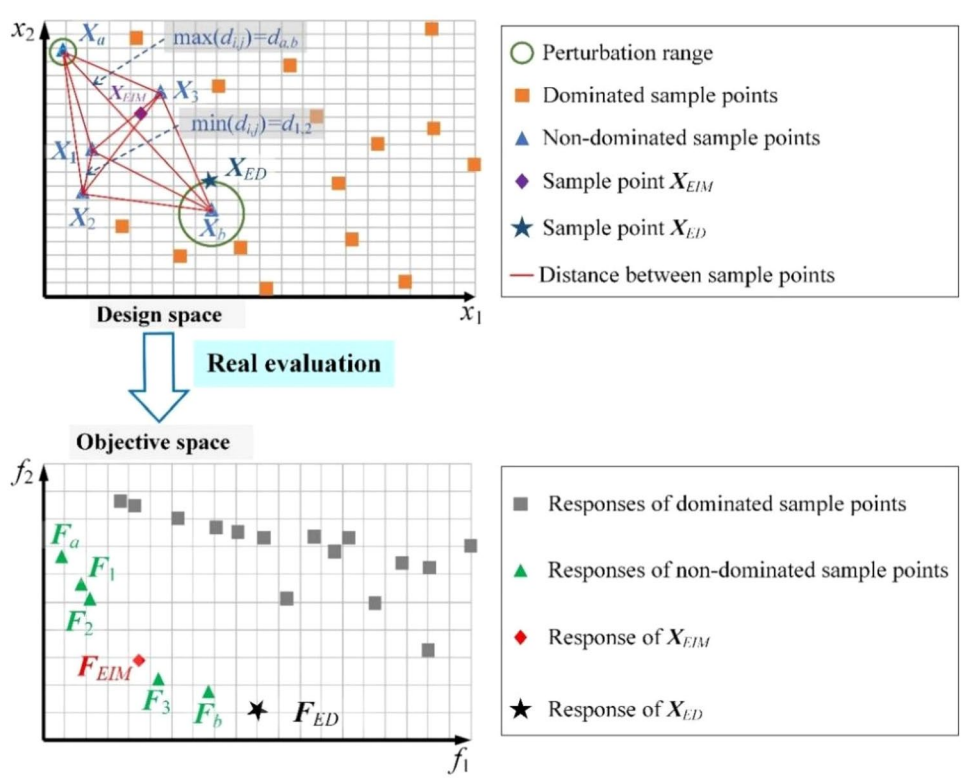


Fig. 2 Diagram of generating new samples by the ED-based infill criterion

where $\delta = \begin{cases} 1-(2r)^{\frac{1}{N_s+1}}, 0 < r \le 0.5 \\ 1-(2(1-r))^{\frac{1}{N_s+1}}, 1 > r > 0.5 \end{cases}$, r is a random number.

According to Eq. (12), Euclidean distances between non-dominated sample points can be evaluated, and the maximum Euler distance $d_{\max}$ in Fig. 2 is related to a pair of sample points X_a and X_b. Then, the perturbation around X_a and X_b is implemented to generate two sample points $X_{p,1}$ and $X_{p,2}$ by the simulated binary crossover method (Agrawal and Deb 1995), where the corresponding kth elements of $X_{p,1}$ and $X_{p,2}$ are evaluated by

$$\begin{cases} X_{p,1,k} = \frac{1}{2}\left[(1-\beta_k)X_{a,k} + (1+\beta_k)X_{b,k}\right] \\ X_{p,2,k} = \frac{1}{2}\left[(1+\beta_k)X_{a,k} + (1-\beta_k)X_{b,k}\right] \end{cases}, \tag{14}$$

where $k = 1, 2, \ldots, M$. Herein, the factor of the simulated binary crossover β_k is defined by

$$\beta_k = \begin{cases} (2t)^{\frac{1}{N_{nds}+1}}, 0 < t \le 0.5 \\ [2(1-t)]^{\frac{-1}{N_{nds}+1}}, 1 > t > 0.5 \end{cases}, \tag{15}$$

where $X_{a,k}$ and $X_{b,k}$, are the kth elements of X_a and X_b, respectively. t is a random number.

Furthermore, the minimum Euclidean distances $d_{\min,1}$ and $d_{\min,2}$ between $X_{p,1}$, $X_{p,2}$ and the sample points are evaluated by

$$\begin{cases} d_{\min,1} = \min\limits_{i=1}^{Ns}\left(\sqrt{\sum_{k=1}^{M}\left(X_{p,1,k} - x_{i,k}\right)^2}\right) \\ d_{\min,2} = \min\limits_{i=1}^{Ns}\left(\sqrt{\sum_{k=1}^{M}\left(X_{p,2,k} - x_{i,k}\right)^2}\right) \end{cases}, \tag{16}$$

By comparing $d_{\min,1}$ and $d_{\min,2}$ with d_p, the infill sample point X_p is expressed as

$$X_p = \begin{cases} \frac{1}{2}\left[tX_{p,1} + (1+t)X_{p,2}\right] \text{ if } d_{\min-1} \ge d_p \text{ and } d_{\min-2} \ge d_p \\ X_{p,1} \text{ if } d_{\min-1} \ge d_p \text{ and } d_{\min-2} < d_p \\ X_{p,2} \text{ if } d_{\min-1} < d_p \text{ and } d_{\min-2} < d_p \end{cases}, \tag{17}$$

and its response F_p is obtained by evaluating the real model.

When the requirement of the Eq. (17) can't be satisfied, the perturbed mechanism with the SBC method will be invoked by Eq. (14). In practical, if the count of the perturbed mechanism invoking reaches a specified threshold, X_p will be evaluated by

$$X_p = tX_a + (1-t)X_b, \tag{18}$$

Moreover, if the element of X_p overflows its boundary, it will be dragged into the reasonable value by

$$X_{p,k} = \begin{cases} (1+t)x_k^L \text{ if } X_{p,k} < x_k^L \\ (1-t)x_k^U \text{ if } X_{p,k} < x_k^U \end{cases}, \tag{19}$$

where x_k^U is the kth element of X_p, x_k^L and x_k^U are the upper and lower boundaries, respectively.

3.3 Filter strategy for overlapped sample points

In this paper, when X_{EIM} and X_p satisfies the requirement of the Eq. (20), they will be regarded as the overlapping sample points

$$\sum_{k=1}^{M}\left|X_{EIM,k} - X_{p,k}\right| \le \Delta, \tag{20}$$

where $X_{EIM,k}$ is the kth element of X_{EIM}. Herein, the threshold Δ is set to 10^{-2}.

Since the overlapping sample points likely lead to the failure of constructing surrogate models, it is necessary to eliminate the repetition phenomenon. Thus, once the infill sample points overlap, the ED-based infill criterion will be employed to generate a new X_p.

3.4 Hybrid criterion-based sample infilling strategy for SMOO

With integrating the EIM_h- and ED-based infill criteria, the hybrid criterion-based sample infilling strategy (HCSIS) is proposed in place of the regular sample infilling format in SMOO. As shown in Fig. 3, at the beginning of each optimization iteration, the current non-dominated solutions are obtained by the differential evolution algorithm in this paper, then they are used as a guide to independently explore the objective space and the design space by the EIM_h- and the ED- based infill criteria respectively, and finally two infill sample points can be generated. Specifically, the infill sample point generated by EIM_h-based infill criterion is used to improve the distribution and convergence of non-dominated solutions in objective space, while the infill sample point generated by ED-based infill criterion is used to exploit the potential areas in design space. Moreover, to ensure the effectiveness of the infill sample points at each iteration, the sample point generated by the ED-based infill criterion will be regenerated through the filter strategy when

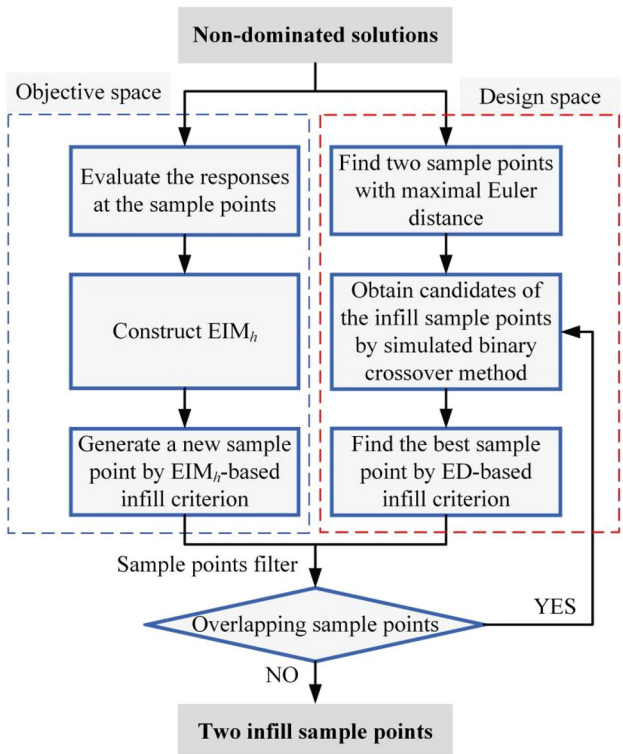


Fig. 3 Framework of the HCSIS-SMOO

it is overlapped with the sample point generated by the EIM$_h$-based infill criterion.

3.5 Computational consumption analysis of HCSIS-SMOO

As shown in Figs. 1 and 3, the computational cost T_{com} of the HCSIS-SMOO is expressed as

$$T_{\text{com}} = N_{\max}\left(\sum_{1}^{N} T_i^{\text{obj}} + \sum_{j=1}^{J+K} T_j^{\text{con}}\right) + N_{iter}\left(\sum_{1}^{N} Ts_i^{\text{obj}} + \sum_{j=1}^{J+K} Ts_j^{\text{con}} + T_{\text{acq}} + T_{\text{o}}\right), \tag{21}$$

where T_i^{obj} and T_j^{con} are the real evaluated costs of ith objective and jth constraint, respectively. Ts_i^{obj} and Ts_j^{con} are the computational costs of constructing surrogate models corresponding to ith objective and jth constraint, respectively. T_{acq} is the computational cost of acquiring sample points based on hybrid infill criterion. T_{o} is the computational cost of assessing feasible constraints and non-dominated solutions in optimization process, and it's usually negligible.

For expensive multi-objective optimization problems, the most time-consuming parts are T_i^{obj} and T_j^{con} which mainly depend on the complexity of the real models. While Ts_i^{obj} and Ts_j^{con} are the less time-consuming parts which mainly depend on the size of experimental samples and the complexity of surrogate models. T_{acq} mainly depends on the cost of gaining sample points by the p-MDED criterion. Generally, T_{acq} is much less than T_i^{obj} and T_j^{con}. Overall, for reducing T_{com} as much as possible, it is vital to control the increase of T_i^{obj} and T_j^{con}. Meanwhile, Ts_i^{obj}, Ts_j^{con} and T_{acq} will also be decreased.

Table 1 Settings for solving test instances with HCSIS-SMOO

Symbol	N_{ini}	N_{iter}	$N_{\max}$	N_T	P_{ref}
Value for ZDTs	80	50	100	500	(1.1,1.1)
Value for Nowacki beam problem	100	50	100	500	(1.5×10^{-2} m^2, 240 MPa)

4 Numerical examples

Since the hypervolume (HV) indicator reveals how close proximity between the non-dominated solutions and true Pareto front (Zitzler and Thiele 1999), and the inverted generational distance (IGD) embodies the distribution and convergence of non-dominated solutions, they are commonly used to evaluate the performance of compared algorithms (Agrawal and Deb 1995). Herein, ZDT instances (ZDTs) (Deb 1999) and Nowacki beam problem (Nowacki 1980) are conducted to validate the feasibility of proposed method. Furthermore, the comparisons of related HV and IGD values are carried out to illustrate the capability of HCSIS-SMOO.

For the purposes of comparisons, the settings of the related parameters are listed in Table 1, where N_{ini} is the number of the initial experimental samples, $N_{\max}$ is the number of the maximum real evaluations, N_{iter} is the number of the maximum iterations, N_T is the number of invoking perturbed mechanism in ED-based infill criterion, P_{ref} is the reference point.

4.1 Ablation experiment

According to the framework of the HCSIS-SMOO, an additional infill sample can be obtained by the ED-based infill criterion except generating an infill sample by the EIM$_h$-based infill criterion. To demonstrate the effectiveness, the ablation experiment is designed in the subsection, and the unconstrained ZDTs are employed to validate

Table 2 Comparisons of HV statistical values between the EIM_h-SMOO and HCSIS-SMOO

Benchmark	Dimension	EIM_h-SMOO				HCSIS-SMOO			
		Min	Max	Mean	s.t.d	Min	Max	Mean	s.t.d
ZDT1	8	0.7239	0.7261	0.7243	0.0029	**0.7288**	**0.7314**	**0.7304**	**0.0023**
	20	0.7218	0.7250	0.7238	0.0041	**0.7282**	**0.7295**	**0.7293**	**0.0033**
	30	0.7206	0.7237	0.7228	0.0049	**0.7280**	**0.7291**	**0.7289**	**0.0038**
ZDT2	8	0.6291	0.6349	0.6340	0.0024	**0.6347**	**0.6436**	**0.6410**	**0.0026**
	20	0.6280	0.6339	0.6325	0.0025	**0.6364**	**0.6445**	**0.6417**	**0.0026**
	30	0.6237	0.6332	0.6313	0.0029	**0.6346**	**0.6442**	**0.6416**	**0.0027**
ZDT3	8	0.7670	1.0406	0.9467	0.1910	**1.0667**	**1.1428**	**1.1254**	**0.0144**
	20	0.5141	0.7990	0.7178	0.2191	**1.0303**	**1.1197**	**1.1088**	**0.0660**
	30	0.1977	0.6613	0.5104	0.3510	**0.6444**	**0.9902**	**0.9814**	**0.1847**
ZDT4	8	0.1124	0.2560	0.2393	**0.1034**	**0.6897**	**0.8895**	**0.8102**	0.1358
	20	0.1107	0.1370	0.0788	0.0749	**0.1325**	**0.1614**	**0.1602**	**0.0283**
ZDT6	8	0.1125	0.1210	0.1161	**0.00401**	**0.3274**	**0.5121**	**0.5076**	0.1508
	20	0.1102	0.1189	0.1108	**0.0923**	**0.2653**	**0.5161**	**0.5024**	0.1985

Bold values indicate the best results

the competitive HVs of HCSIS-SMOO. Herein, 10 groups initial experimental samples are selected for solving multi-objective optimization problems, and each optimization cycle runs 20 times independently.

As Table 2 lists, HCSIS-SMOO integrating two infill criteria obtained larger and more stable HV than EIM_h-SMOO (Zhan et al. 2017) including only one infilling criterion under the same conditions. Since the larger HV shows the closer non-dominated solutions to the true Pareto front (Zitzler and Thiele 1999), the HCSIS can help the SMOO obtain better non-dominated solutions than the EIM_h-based infill criterion. This is because the HCSIS-SMOO not only retains the advantages of EIM_h-SMOO in exploring objective space, but also can obtain more diverse infill samples by exploiting the design space on this basis, thus promoting the performance of SMOO algorithm.

4.2 Numerical validation on unconstrained multi-objective optimization problems

The ZDTs are still employed to further validate the performance of HCSIS-SMOO, and the TIC-SMEA (-WS and -WMO) (Li et al. 2021a, b), MOEA/D-EGO (Zhang et al. 2010), KRVEA (Chugh et al. 2018) and EIM-EGO (Zhan et al. 2017) are chosen as compared algorithms. The initial experimental samples and the number of the independent optimization cycles are the same as the Sect. 4.1.

With the HCSIS-SMOO and compared algorithms, the statistical results of IGD values for ZDTs are listed in Table 3. The compared results related to compared algorithms are referenced from (Li et al. 2021a, b), '(.)' denotes standard deviation value. Towards more reasonable statistical results, the Wilcoxon rank-sum test calculated at a significance level of $\alpha = 0.05$ is used to test these algorithms (Derrac et al. 2011). For clear explanation, '*' indicates the best result, '≈' represents no statistically significant difference, '+' represents significantly better performance, while '−' represents less as good as others. Moreover, the best mean values and standard deviation values are highlighted.

In principle, the proposed HCSIS-SMOO picks the improved trends of Pareto front in objective space by EIM_h-based criterion and captures the promising area around non-dominated solutions in design space by ED-based criterion. Thus, it can generate refined infill samples and further promote the diversity of DNSs, whereby improving the performance of IGD. As listed in Table 3, as for the mean IGD values obtained by HCSIS-SMOO, despite of none statistically differences on 8-dimension ZDT1, 8-dimension ZDT2 and 30-dimension ZDT6, the rest are significantly better than other compared algorithms. It indicates that the HCSIS-SMOO has enhanced the quality of IGD, accordingly, the performances (distribution and convergence) of non-dominated solutions can also be improved.

Essentially, since the current non-dominated solutions are referred to generate promising infill samples, the usefully known information is transferred to next optimization iteration. Ultimately, despite of the influences with different groups of initial samples, the non-dominated solutions closest to true Pareto front will be likely to be find out. Correspondingly, the fluctuation of IGD values will also be kept in a small range. As shown in Table 3, for standard deviations of IGD values obtained by HCSIS-SMOO,

Table 3 Comparisons of IGD statistical values (mean values and standard deviations)

Benchmark	Dimension	HCSIS-SMOO	TIC-SMEA	TIC-SMEA-WS	TIC-SMEA-WMO	MOEA/D-EGO	KRVEA	EIM-EGO
ZDT1	8	**0.00483≈**	**0.00549***	0.01014	0.00641	0.04334	0.3929	0.40787
		(0.00024) +	**(0.00071)***	(0.0029)	(0.00083)	(0.02056)	(0.01702)	(0.2569)
	20	**0.00488 +**	**0.00892***	0.01026	0.01513	34.17603	4.53869	24.1037
		(0.00021) +	**(0.00126)***	(0.0021)	(0.00312)	(7.6627)	(8.86177)	(9.02568)
	30	**0.00545 +**	**0.01121***	0.01326	0.04907	74.7784	75.4793	68.03469
		(0.00056) +	**(0.00272)***	(0.00373)	(0.00312)	(10.10952)	(24.63102)	(6.96045)
ZDT2	8	**0.00584 ≈**	**0.00691***	0.00829	0.00741	0.06043	0.04909	0.52323
		(0.00045) ≈	**(0.00051) ***	(0.00072)	(0.00088)	(0.04043)	(0.01705)	(0.12444)
	20	**0.00512 +**	**0.00768***	0.0131	0.6662	36.5725	14.31183	30.7056
		(0.00056) +	**(0.00148)***	(0.0086)	(0.05846)	(7.28628)	(16.25344)	(6.84603)
	30	**0.00579 +**	**0.01235***	0.02463	0.15477	74.68902	92.16315	63.34392
		(0.00103) +	**(0.01249)***	(0.02448)	(0.07237)	(16.76366)	(14.39516)	(14.53882)
ZDT3	8	**0.03182 +**	0.08755	0.0625	0.14389	0.16501	**0.05638***	0.4567
		(0.00903) +	(0.04602)	(0.03622)	(0.08702)	(0.06702)	**(0.02932)***	(0.17253)
	20	**0.06202 +**	0.1269	**0.08319***	0.1943	38.00012	11.32679	25.02083
		(0.03895) ≈	(0.05367)	**(0.03532)***	(0.10639)	(7.29328)	(18.87377)	(6.4064)
	30	**0.09802 +**	**0.11238***	0.14901	0.19515	72.85005	77.01886	63.23996
		(0.01555) +	**(0.05269)***	(0.09787)	(0.08068)	(7.69604)	(22.59039)	(8.21681)
ZDT4	8	**0.10689 +**	19.75013	**16.78597***	20.81498	58.35555	22.08141	36.75476
		(0.12569) +	(14.93557)	**(8.27995)***	(8.82016)	(10.1773)	(9.03379)	(9.64027)
	20	**0.38614 +**	117.99853	**105.46418***	134.86659	225.91427	168.43922	183.6275
		(0.08957) +	(37.7449)	**(21.5474)***	(24.09434)	(13.57934)	(29.92508)	(16.44542)
	30	**1.08916 +**	178.03733	**170.53677***	260.51451	356.77022	319.74428	257.10737
		(0.42447) +	(38.21438)	**(36.55844)***	(35.66436)	(28.18849)	(33.78974)	(37.46149)
ZDT6	8	**0.0465 +**	0.34431	**0.29901***	0.68054	0.87704	1.57076	0.60435
		(0.02199) +	(0.16958)	**(0.10623)***	(0.37856)	(0.89574)	(0.37484)	(0.57468)
	20	**0.12722 +**	**0.46785***	0.47859	5.49001	13.39675	8.57976	7.02407
		(0.04346) ≈	**(0.04569)***	(0.04512)	(0.7074)	(0.75211)	(2.3564)	(1.29681)
	30	**0.31514 ≈**	0.51136	**0.47353***	9.22935	15.99963	15.79114	11.81294
		(0.50753) −	(0.08682)	**(0.07688)***	(1.05887)	(0.5256)	(0.91364)	(1.18859)

Bold values indicate the best results

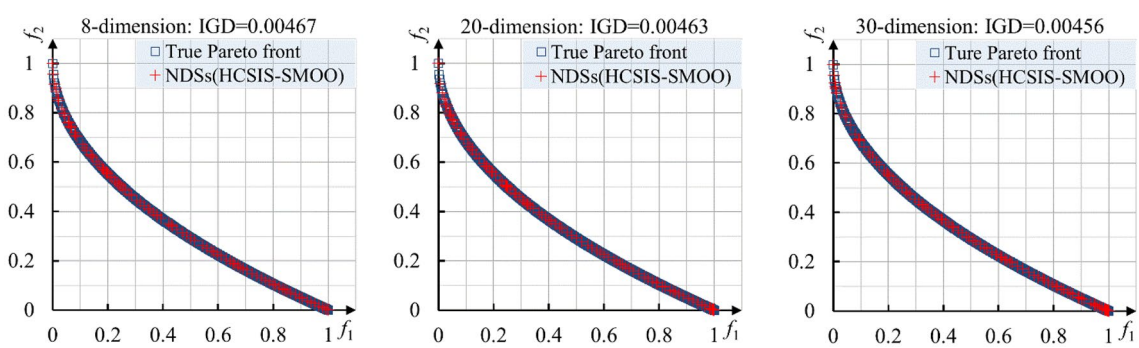


Fig. 4 Non-dominated solutions with lowest IGD values for ZDT1

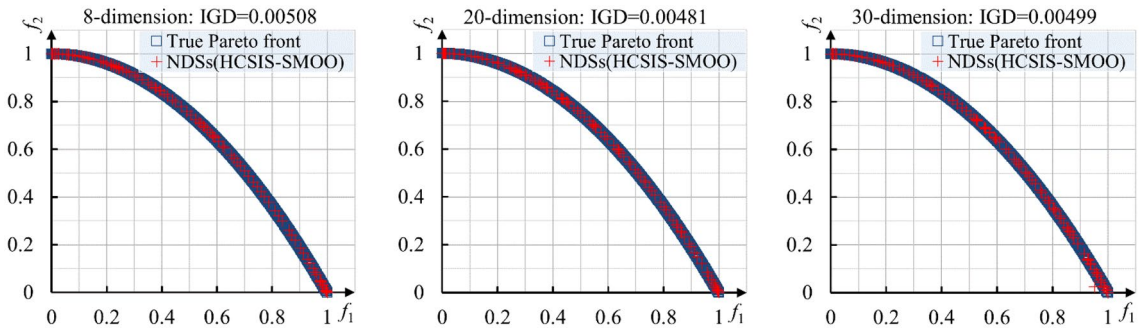


Fig. 5 Non-dominated solutions with lowest IGD values for ZDT2

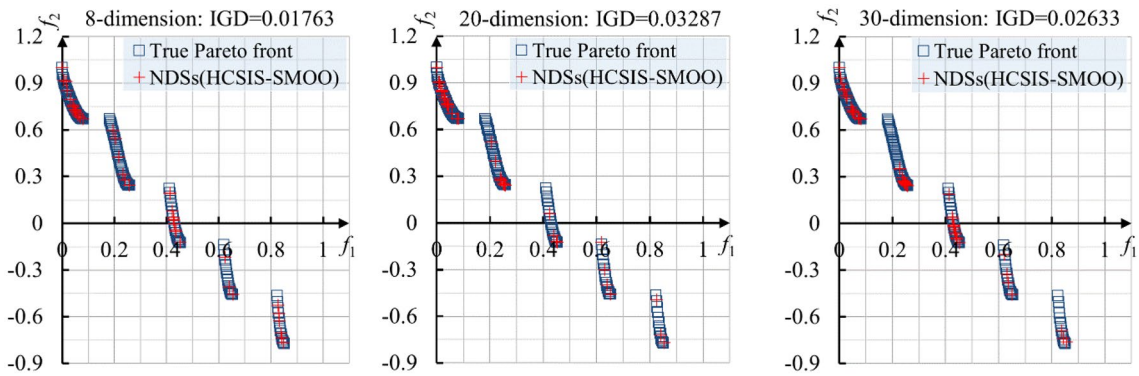


Fig. 6 Non-dominated solutions with lowest IGD values for ZDT3

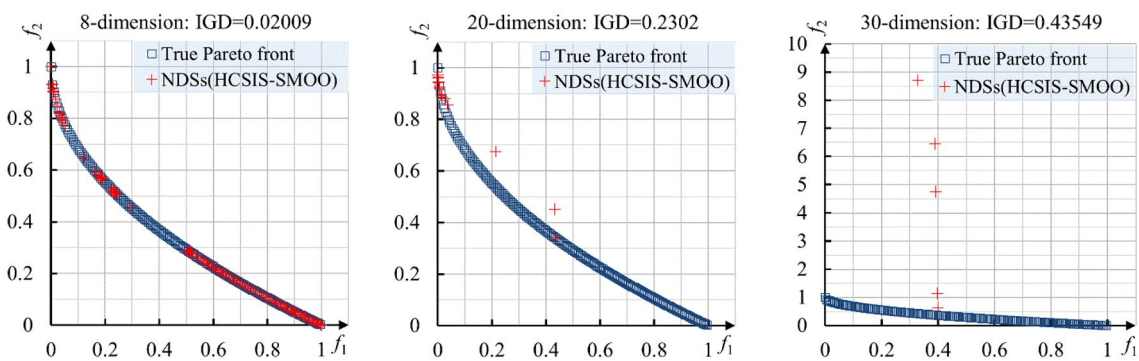


Fig. 7 Non-dominated solutions with lowest IGD values for ZDT4

most of them are advantageous over others, although there are none statistically significant differences on 8-dimension ZDT2, 20-dimension ZDT3 and 20-dimension ZDT6. For 30-dimension ZDT6, the performance of HCSIS-SMOO is poorer than that of TIC-SMEA, however, the overall results listed in Table 3 still show the competitive computational stability of the proposed algorithm.

To further display the results obtained by HCSIS-SMOO, Figs. 4, 5, 6, 7 and 8 show the final non-dominated solutions with lowest IGD values of the ZDTs. For ZDT1 and ZDT2 with continuous Pareto front, the non-dominated solutions for each group match well with the true Pareto front and distributes evenly, and the same is for non-dominated solutions of ZDT3 with discontinuous Pareto front. When solving an expensive multi-objective optimization problem by MOEA, the stronger multi-peak characteristic probably

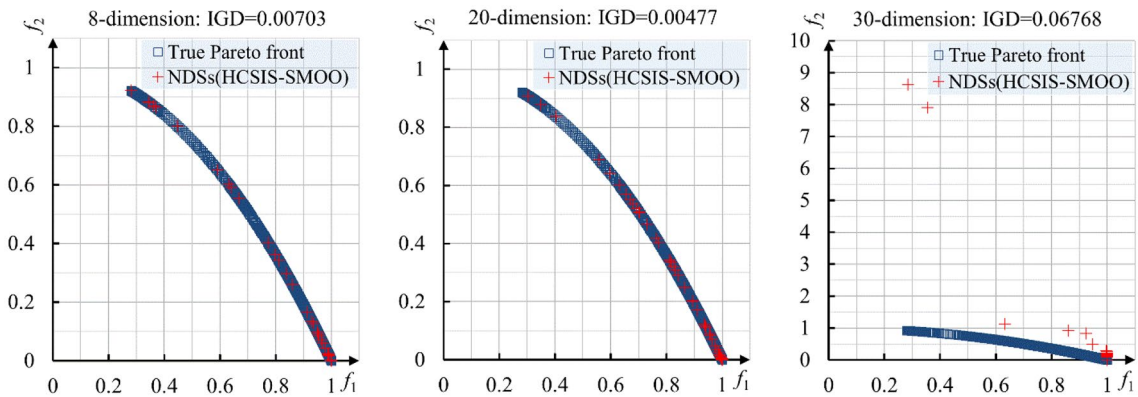


Fig. 8 Non-dominated solutions with lowest IGD values for ZDT6

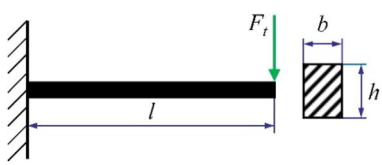


Fig. 9 Structural parameters of Nowacki beam

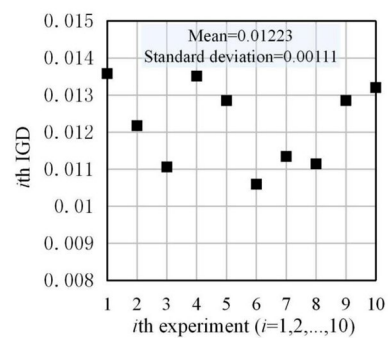


Fig. 10 IGD values of 10 groups non-dominated solutions for Nowacki beam problem

leads to poorer global solutions. Thus, as for ZDT4 and ZDT6 with continuous Pareto front, due to the stronger multi-peak characteristics, the performances of non-dominated solutions are poorer than those of ZDT1-ZDT3. Even so, the non-dominated solutions for each group under low-dimension conditions still well match with the true Pareto front. Generally, it indicates the considerable capacity of HCSIS-SMOO in addressing the multi-peak multi-objective optimization problems.

4.3 Numerical validation on constrained multi-objective optimization problem

The Nowacki beam problem is a constrained multi-objective optimization problem related to a cantilever beam (Fig. 9), in which, minimizing the cross-sectional area A and bending stress σ_B are regarded as the optimization objectives constrained by structural performances (Nowacki 1980).

The Nowacki beam problem is expressed as

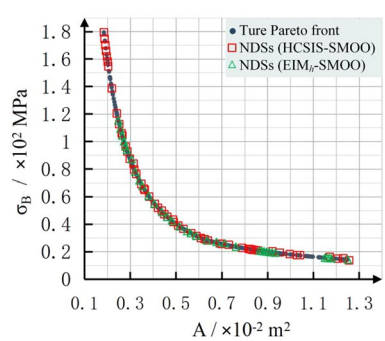


Fig. 11 Comparison of non-dominated solutions with lowest IGD by HCSIS- and EIM$_h$- SMOO in solving Nowacki beam problem

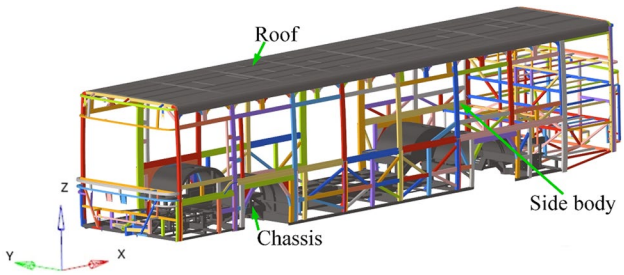


Fig. 12 Composition of electric bus body

$$\begin{cases} \text{find } [h, b] \\ \text{Min } [A = hb, \sigma_B = \dfrac{6F_t l}{bh^2}] \\ s.t. \begin{cases} \sigma_{\text{beam}} \le 5 \text{ mm} \\ \sigma_B \le \sigma_Y \\ 2\tau \le \sigma_Y \\ h \le 10b \\ f_s \times F_t \le F_{cr} \\ 10 \text{ mm} \le h \le 50 \text{ mm}, 20 \text{ mm} \le b \le 250 \text{ mm} \end{cases} \end{cases}, \quad (22)$$

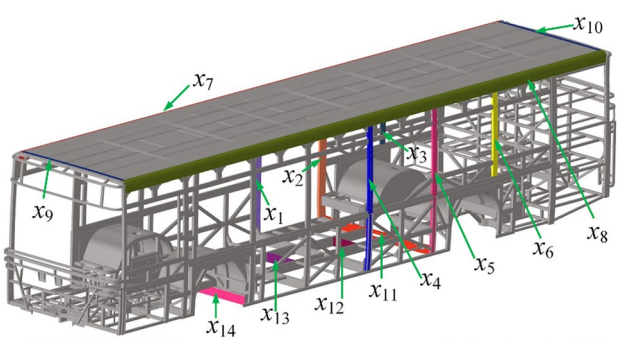


Fig. 13 Optimized objects and design variables of electric bus body

where l denotes the length of the cantilever beam, h denotes the height of the cross section, b denotes the width of the cross section. F_t denotes the load on the tip of the cantilever beam. δ_{beam} denotes the corresponding tip deflection, σ_Y denotes the yield stress, τ denotes the shear stress, F_{cr} denotes the buckling failure force, and f_s denotes the safety factor. By referring to the Ref. (Zhan et al. 2017), l is set to 1.5 m, F_t is set to 5 kN.

To address the Nowacki beam problem, the probability of feasibility criterion is introduced to HCSIS-SMOO (Parr et al. 2012). The ED-based infill criterion with constraints is expressed as

$$\text{PoF}(\mathbf{x}) = \prod_{j=1}^{J} \left[\Phi\left(\frac{0 - \hat{g}_j(\mathbf{x})}{s_{g,j}(\mathbf{x})} \right) \right], \quad (23)$$

where $\text{X} = [x_1, x_2, \ldots, X_{14}]$ is the Kriging model predicted value of the jth constraint, $s_{g,j}(\mathbf{x})$ is the square root mean error of the jth constraint. While the EIM_h-based infill criterion with constraints is expressed as

$$\begin{aligned} \text{C - EIM}_h(\mathbf{x}) = & \min_{j=1}^{N_{nds}} \left(\prod_{i=1}^{N} \left(r_i + \text{EI}_i^j(\mathbf{x}) - f_i^j \right) - \prod_{i=1}^{N} \left(r_i - f_i^j \right) \right) \\ & \times \prod_{j=1}^{J} \left(\Phi\left(\frac{0 - \hat{g}_j(\mathbf{x})}{s_{g,j}(\mathbf{x})} \right) \right), \end{aligned} \quad (24)$$

10 groups of initial experimental samples are selected for validation, and other settings are listed in Table 1. As a result, 10 groups IGD and non-dominated solutions with lowest IGD (0.0106) are shown in Figs. 10 and 11, respectively. Moreover, the non-dominated solutions with the lowest IGD (0.0112) obtained by EIM_h-SMOO are also displayed in Fig. 11, and the comparison results show that the non-dominated solutions obtained by the HCSIS-SMOO have advantages in matching with the true Pareto front and the counts of the non-dominated solutions, and it further indicates that HCSIS-SMOO has competitive ability when solving the constrained multi-objective optimization problem.

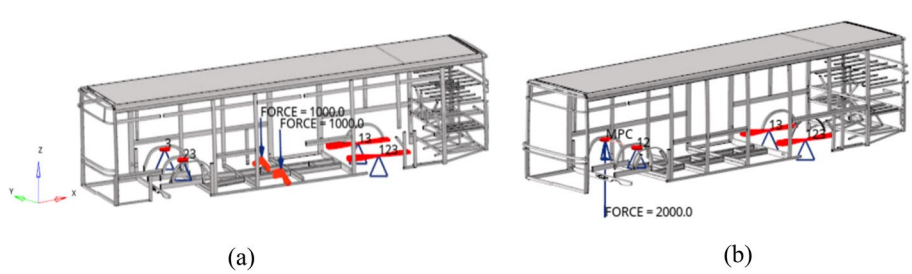


Fig. 14 Forces and constraints on electric bus body. **a** bending case; **b** torsional case

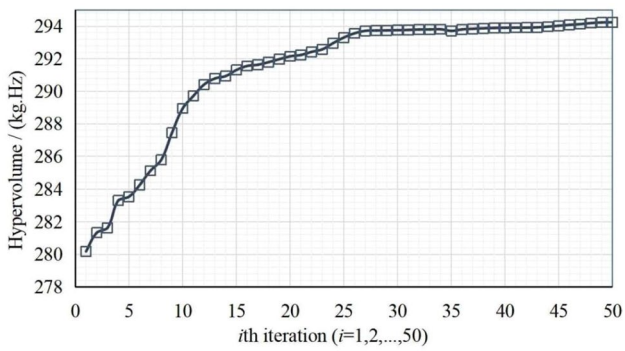


Fig. 15 HV trace of solving lightweight design problem of electric bus body structure

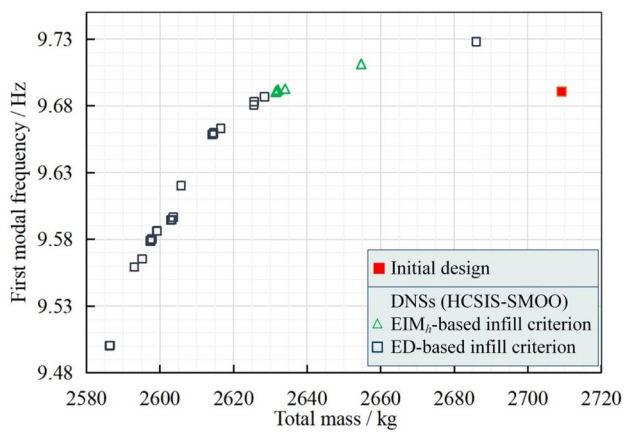


Fig. 16 Lightweight design results of electric bus body structure

5 Lightweight design of electric bus body

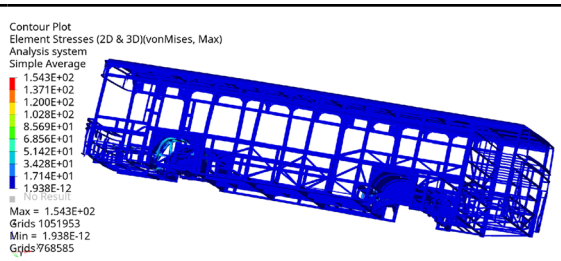
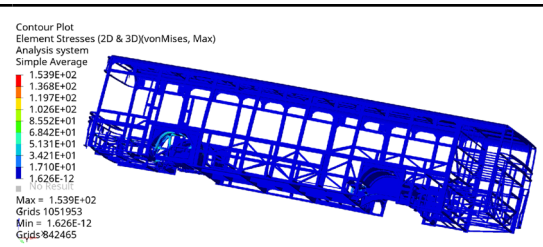
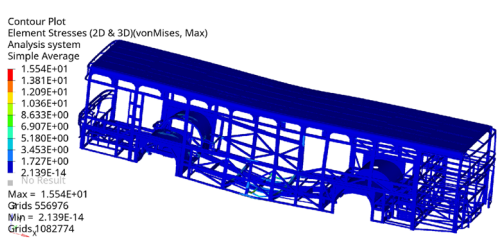
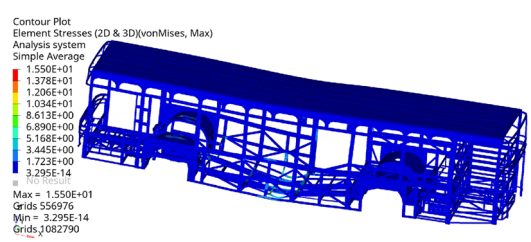
Due to constraints of power battery energy density, the lightweight of the electric bus body structure is attracting more attentions for the mileage extending (Hannan et al. 2014). For the multi-material electric bus body mentioned in Fu et al. (2019), the proposed method is employed to realize the lightweight design with high comprehensive performances.

As shown in Fig. 12, the multi-material electric bus body contains three substructures which are the side body, the roof and the chassis, respectively. The finite element model in this study has been validated in Fu et al. (2019), which contains 1,802,597 two-dimension elements.

The design variables and structural materials are referred to Fu et al. (2019). As Fig. 13 shows, the thicknesses of fourteen parts are considered as design variables that are represented by the vector $X = [x_1, x_2, \ldots, x_{14}]$, where the initial values of $x_1, x_2, \ldots, x_{10}$ are set to 4 mm, the corresponding material is aluminum. While the initial values of x_{11}, x_{12}, x_{13}, x_{14} are 5 mm, the corresponding material is steel. Moreover, Young's modulus, Poisson's ratio and density of the aluminum and the steel are set to 69 GPa and 210 GPa, 0.33 and 0.3, 2700 kg/m^3 and 7800 kg/m^3, respectively.

Towards lightweight design with high performances, minimizing the total mass and maximizing the first order modal frequency are regarded as the optimization objectives constrained by the bending and torsional stiffness, then the multi-objective optimization model is expressed as

Table 4 Comparisons of structural performances before and after optimization

	Initial design	HCSIS-SMOO
Torsional stress		
Bending stress		

$$\begin{cases} \text{Find } \mathbf{X} = [x_1, x_2, \ \dots \ , x_{14}] \\ \text{Min } \ [m_t, -f_{freq}] \\ s.\ t. \begin{cases} d_b \le 3.6 \text{ mm} \\ d_t \le 1.8 \text{ mm} \\ 2 \text{ mm} \le x_k \le 6 \text{ mm}, k = 1, 2, \dots 14 \end{cases} \end{cases}, \tag{25}$$

where m_t is the total mass, f_{freq} is the first order modal frequency. d_b and d_t are maximum Z-axis displacement under bending case and torsional case, respectively. The above performances are evaluated through finite element analysis.

Loads and constraints involved under the two static cases are shown in Fig. 14, in which, the unit of force is "N", the number "1, 2, 3" represent that the translation motion along the X, Y and Z axis are constrained, respectively. According to the Table 1, settings of solving this problem are defined as follows,$N_{\text{ini}} = 200$, $N_{\text{iter}} = 50$, $N_{\max} = 100$, $N_T = 500$. $\text{P}_{\text{ref}} = \left(m_t^{\text{ref}}, f_{\text{freq}}^{\text{ref}}\right)$, where the referred total mass m_t^{ref} is set to 3000 kg, the referred first order modal frequency $f_{\text{freq}}^{\text{ref}}$ is set to 9 Hz.

After 50 optimization iterations, the HV indicator trace and 31 non-dominated solutions obtained by the HCSIS-SMOO are shown in Figs. 15 and 16, respectively.

As Fig. 15 shows, the value of HV is basically stable after the 27th iteration, and the relative error is less than 0.15%. As Fig. 16 shows, only 8 non-dominated solutions originate from the EIM_h-based infill criterion, while 23 non-dominated solutions originate from the DE-based infill criterion. Therefore, more diverse non-dominated solutions have been obtained by the HCSIS-SMOO.

Compared with the initial design (2709 kg, 9.69 Hz), all the optimization results have advantages in lightweight, and two of them have obvious advantages in first order modal frequency. Herein, the result (2686 kg, 9.73 Hz) with the maximum first order modal frequency is employed to further show the comparisons of the structural performances. As listed in Table 4, with the weight reduction (23 kg) and the frequency increase (0.04 Hz), the maximum structural stress of the torsional case decreases from 154.3 to 153.9 MPa, and the maximum structural stress of the bending case decreases from 155.4 to 155 MPa. It indicates that the present optimization solution can achieve light weight while maintaining the strength performances of the electric bus body structure.

6 Conclusions

In this paper, a hybrid criterion-based sample infilling strategy for surrogate-assisted multi-objective optimization is proposed, by which two promising samples can be obtained to update the surrogate models in every SMOO iteration, with this help to improve the predicted performance of surrogate model and promote the multi-objective optimization results. Through the numerical validations with well-known benchmarks and performance comparisons with several advanced algorithm, it demonstrates that the HCSIS-SMOO has competitive advantages in obtaining high-quality Pareto front and computational stability. On the above basis, the proposed method is employed to realize the lightweight design of an electric bus body structure, and several higher-performance design solutions are obtained for decision-makers. Generally, the HCSIS-SMOO method is verified in solving high-dimensional multi-objective optimization engineering problems.

Acknowledgements This work is supported by the National Natural Science Foundation of China (Grant No. 51805032).

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Replication of results The codes of EIM_h algorithm based on reinforcement learning algorithm can be founded at https://github.com/zhandawei/Efficient_Global_Optimization_Algorithms. All the datasets generated in this work are available upon reasonable request to the corresponding author.

References

Agrawal R, Deb K (1995) Simulated binary crossover for continuous search space. Complex Syst 9(2):115–148

Bader J, Zitzler E (2011) HypE: an algorithm for fast hypervolume-based many-objective optimization. Evolut Comput 19(1):45–76. https://doi.org/10.1162/evco_a_00009

Chen ZS, Hou KR, Zhu MY, Xu Y, Zhu QX (2021) A virtual sample generation approach based on a modified conditional GAN and centroidal Voronoi tessellation sampling to cope with small sample size problems application to soft sensing for chemical process. Appl Soft Comput 101:107070. https://doi.org/10.1016/j.asoc.2020.107070

Cheng J, Jiang P, Zhou Q, Hu JX, Yu T, Shu LS, Shao XY (2019) A lower confidence bounding approach based on the coefficient of variation for expensive global design optimization. Eng Comput 36(3):830–849. https://doi.org/10.1108/EC-08-2018-0390

Chugh T, Jin YC, Miettinen K, Hakanen J, Sindhya K (2018) A surrogate-assisted reference vector guided evolutionary algorithm for computationally expensive many-objective optimization. IEEE Trans Evol Comput 22(1):129–142. https://doi.org/10.1109/TEVC.2016.2622301

Chugh T, Sindhya K, Hakanen J, Miettinen K (2019) A survey on handling computationally expensive multi-objective optimization problems with evolutionary algorithms. Soft Comput 23(9):3137–3166. https://doi.org/10.1007/s00500-017-2965-0

Couckuyt I, Deschrijver D, Dhaene T (2014) Fast ccalculation of multi-objective probability of improvement and expected improvement criteria for Pareto optimization. J Glob Optim 60(3):575–594. https://doi.org/10.1007/s10898-013-0118-2

Deb K (1999) Multi-objective genetic algorithms: problem difficulties and construction of test problems. Evol Comput 7(3):205–230. https://doi.org/10.1162/evco.1999.7.3.205

Derrac J, García S, Molina D, Herrera F (2011) A practical tutorial on the use of non-parametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. Swarm Evol Comput 1(1):3–18. https://doi.org/10.1016/j.swevo.2011.02.002

Dong HC, Sun SQ, Song BW, Wang P (2019) Multi-surrogate-based global optimization using a score-based infill criterion. Struct Multidisc Optim 59(2):485–506. https://doi.org/10.1007/s00158-018-2079-z

Fu CL, Bai YC, Lin C, Wang WW (2019) Design optimization of a newly developed aluminum-steel multi-material electric bus body structure. Struct Multidisc Optim 60(5):2177–2187. https://doi.org/10.1007/s00158-019-02292-w

Gonzalez S, Jalali H, Inneke V (2020) A multi-objective stochastic simulation optimization algorithm. Eur J Oper Res 284(1):212–226. https://doi.org/10.1016/j.ejor.2019.12.014

Guo ZD, Wang QN, Song LM, Li J (2021) Parallel multi-fidelity expected improvement method for efficient global optimization. Struct Multidisc Optim 64(3):1457–1468. https://doi.org/10.1007/s00158-021-02931-1

Hannan M, Azidin F, Mohamed A (2014) Hybrid electric vehicles and their challenges: a review. Renew Sustain Energy Rev 29:135–150. https://doi.org/10.1016/j.rser.2013.08.097

He YW, Sun JJ, Song P, Wang XS (2021) Multi-objective efficient global optimization of expensive simulation-based problem in presence of simulation failures. Eng Comput 124(1):1–26. https://doi.org/10.1007/s00366-021-01351-5

Horn D, Wagner T, Biermann D, Weihs C, Bischl B (2015) Model based multi-objective optimization: taxonomy, multi-point proposal, poolbox and benchmark. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 9018:64–78. https://doi.org/10.1007/978-3-319-15934-8_5

Hou SJ, Tan W, Zheng YN, Han X, Li Q (2014) Optimization design of corrugated beam guardrail based on RBF-MQ surrogate model and collision safety consideration. Adv Eng Softw 78:28–40. https://doi.org/10.1016/j.advengsoft.2014.08.002

Jeong S, Minemura Y, Obayashi S (2006) Optimization of combustion chamber for diesel engine using Kriging model. J Fluid Sci Technol 1(2):138–146. https://doi.org/10.1299/jfst.1.138

Jiang Z, Chen SS, Chen W (2016) Reduction of epistemic model uncertainty in simulation-based multidisciplinary design. J Mech Des 138(8):081403. https://doi.org/10.1115/1.4033918

Jin YC (2011) Surrogate-assisted evolutionary computation: recent advances and future challenges. Swarm Evol Comput 1(2):61–70. https://doi.org/10.1016/j.swevo.2011.05.001

Jones D, Schonlau M, Welch W (1998) Efficient global optimization of expensive black-box functions. J Glob Optim 13(4):455–492. https://doi.org/10.1023/A:1008306431147

Klamroth K, Lacour R, Vanderpooten D (2015) On the representation of the search region in multi-objective optimization. Eur J Oper Res 245(3):767–778. https://doi.org/10.1016/j.ejor.2015.03.031

Li DC, Huang WT, Chen CC, Chang CJ (2013) Employing virtual samples to build early high dimensional manufacturing models. Int J Prod Res 51(11):3206–3224. https://doi.org/10.1080/00207543.2012.746795

Li Z, Wang XY, Ruan SL, Li ZJ, Shen CY, Zeng Y (2018) A modified hypervolume based expected improvement for multi-objective efficient global optimization method. Struct Multidisc Optim 58(5):1961–1979. https://doi.org/10.1007/s00158-018-2006-3

Li F, Gao LK, Garg A, Shen WM, Huang SF (2021a) Two infill criteria driven surrogate-assisted multi-objective evolutionary algorithms for computationally expensive problems with medium dimensions. Swarm Evol Comput 60:100774. https://doi.org/10.1016/j.swevo.2020.100774

Li MY, Yi B, Yang Y (2021b) An efficient global optimization method with multi-point infill sampling based on kriging. Eng Optim 54(11):1–18. https://doi.org/10.1080/0305215X.2021.1960985

Liu B, Zhang QF, Gielen G (2014) A Gaussian process surrogate model assisted evolutionary algorithm for medium scale expensive optimization problems. IEEE Trans Evol Comput 18(2):180–192. https://doi.org/10.1109/TEVC.2013.2248012

Namura N, Shimoyama K, Obayashi S (2017) Expected improvement of penalty-based boundary intersection for expensive multi-objective optimization. IEEE Trans Evol Comput 21(6):898–913. https://doi.org/10.1109/TEVC.2017.2693320

Nowacki H (1980) Modelling of design decisions for CAD. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 89(1):177–223. https://doi.org/10.1007/BFb0040161

Pal A, Wang Y, Zhu L, Zhu GG (2021) Multi-objective surrogate-assisted stochastic optimization for engine calibration. J Eng Gas Turbines Power 143(10):101004. https://doi.org/10.1115/1.4050970

Parr J, Keane A, Forrester A, Holden C (2012) Infill sampling criteria for surrogate-based constrained optimization. Eng Optim 44(10):1147–1166. https://doi.org/10.1080/0305215X.2011.637556

Shimoyama K, Sato K, Jeong S, Obayashi S (2013) Updating Kriging surrogate models based on the hypervolume indicator in multi-objective optimization. J Mech Des 135(9):94503–94507. https://doi.org/10.1115/1.4024849

Tang YF, Long T, Shi RH, Wu YF, Wang G-G (2020) Sequential radial basis function-based optimization method using virtual sample generation. J Mech Des 142(11):111701. https://doi.org/10.1115/1.4046650

Wagner T, Emmerich M, Deutz A, Ponweiser W (2010) On expected improvement criteria for model-based multi-objective optimization. Lect Notes Comput Sci 6238(1):718–727. https://doi.org/10.1007/978-3-642-15844-5_72

Wang G-G, Shan SQ (2007) Review of metamodeling techniques in support of engineering design optimization. J Mech Des 129(4):370–380. https://doi.org/10.1115/1.2429697

Wedyan M, Crippa A, Adel A (2019) A novel virtual sample generation method to overcome the small sample size problem in computer aided medical diagnosing. Algorithms 12(8):160. https://doi.org/10.3390/a12080160

Wu YF, Long T, Shi RH, Wang G-G (2021) Mode-pursuing sampling method using discriminative coordinate perturbation for high dimensional expensive black box optimization. J Mech Des 143(4):041703. https://doi.org/10.1115/1.4047909

Xu YW, Renteria A, Wang PF, Jiang PY (2022) Adaptive surrogate models with partially observed information. Reliab Eng Syst Saf 225:108566. https://doi.org/10.1016/j.ress.2022.108566

Yao YM, Yang XD (2021) Efficient global multi-objective aerodynamic optimization using combined multi-point infilling strategy and surrogate models. In: Proceedings of 2021 IEEE congress on evolutionary computation, pp 1537–1542. https://ieeexplore.ieee.org/document/9504946

Yi S, Kwon H, Choi S (2014) Efficient global optimization using a multi-point and multi-objective infill sampling criteria. In: Proceedings of 52nd aerospace sciences meeting, pp 1–19. https://doi.org/10.2514/6.2014-0898

Zhan DW, Cheng YS, Liu J (2017) Expected improvement matrix-based infill criteria for expensive multi-objective optimization. IEEE Trans Evolut Comput 21(6):956–975. https://doi.org/10.1109/TEVC.2017.2697503

Zhang QF, Liu WD, Tsang E, Virginas B (2010) Expensive multi-objective optimization by MOEA/D with Gaussian process model. IEEE Trans Evol Comput 14(3):456–474. https://doi.org/10.1109/TEVC.2009.2033671

Zhang Y, Han ZH, Zhang KS (2019) Variable-fidelity expected improvement method for efficient global optimization of expensive functions. Struct Multidisc Optim 58(4):1431–1451. https://doi.org/10.1007/s00158-018-1971-x

Zhang Y, Wang S, Zhou CA, Lv LY, Song XG (2021) A fast active learning method in design of experiments: multipeak parallel adaptive infilling strategy based on expected improvement. Struct Multidisc Optim 64(3):1259–1284. https://doi.org/10.1007/s00158-021-02915-1

Zitzler E, Thiele L (1999) Multi-objective evolutionary algorithms: a comparative case study and the strength pareto approach. IEEE Trans Evol Comput 3(4):257–271. https://doi.org/10.1109/4235.797969

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.